Electronic Brake Control (EBC) Software — High-Level Design

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Revision History

Revision	Date	Description
A	2026-03-02	Initial architecture baseline.
B	2026-06-18	Added PedalInputHandler, removed HmiWarningManager, updated signal resolutions, defined IF_ActuatorStatus, added degraded mode section.

1 Introduction

This High-Level Design (HLD) describes the software architecture of the Electronic Brake Control (EBC) application layer for a fictional passenger vehicle platform. It identifies the software components, their ports and port interfaces, the exchanged signals, the dependencies between components and the main functional flows. The document is intended for software architects, integration engineers and test engineers.

2 Scope

The scope covers the application software components hosted on the brake ECU. Basic software modules, the hydraulic hardware and calibration data are outside the scope of this document.

3 Architecture Overview

The EBC application is organised as a processing chain. Pedal and wheel sensor information is converted into a vehicle state estimate, which is used to arbitrate a brake torque request. The arbitrated request is executed by the actuator layer. Diagnostic events from all components are collected centrally.

4 Software Components

Table 4-1 lists the application software components (SWCs) and their responsibilities.

ID	Component	Type	Description
C-01	WheelSpeedProcessor	Application SWC	Converts raw wheel sensor pulses into filtered wheel speeds for all four wheels every 10 ms.
C-02	VehicleStateEstimator	Application SWC	Estimates vehicle longitudinal speed and wheel slip from wheel speeds.
C-03	BrakeTorqueCoordinator	Application SWC	Arbitrates brake torque requests and computes the target brake torque per axle.
C-04	BrakeActuatorManager	Application SWC	Commands the hydraulic modulator valves to realise the target brake torque.
C-05	DiagnosticEventHandler	Service SWC	Collects diagnostic events reported by the EBC components and forwards them to the diagnostic stack.
C-06	PedalInputHandler	Sensor-Actuator SWC	Reads and plausibilises the brake pedal travel sensor and provides the driver brake request.

4.1 WheelSpeedProcessor

Responsibilities: filter wheel sensor pulses, detect implausible wheel speed values and provide wheel speeds to downstream components.

The WheelSpeedProcessor sends Sig_WheelSpeed_FL to the VehicleStateEstimator together with the other three wheel speed signals.

4.2 VehicleStateEstimator

The VehicleStateEstimator depends on the WheelSpeedProcessor. It uses Sig_YawRate, when available, to improve the slip estimate in curves.

4.3 BrakeTorqueCoordinator

The BrakeTorqueCoordinator receives Sig_BrakePedalPosition from the PedalInputHandler and combines it with the vehicle state to compute Sig_BrakeTorqueRequest.

4.4 BrakeActuatorManager

The BrakeActuatorManager depends on the BrakeTorqueCoordinator and reports actuator health via the PP_ActuatorStatus port.

5 Ports and Port Interfaces

Component	Port	Direction	Interface	Interface Type
WheelSpeedProcessor	PP_WheelSpeed	Provided	IF_WheelSpeed	Sender-Receiver
VehicleStateEstimator	RP_WheelSpeed	Required	IF_WheelSpeed	Sender-Receiver
VehicleStateEstimator	PP_VehicleState	Provided	IF_VehicleState	Sender-Receiver
BrakeTorqueCoordinator	RP_VehicleState	Required	IF_VehicleState	Sender-Receiver
BrakeTorqueCoordinator	PP_BrakeTorqueCmd	Provided	IF_BrakeTorqueCmd	Sender-Receiver
BrakeActuatorManager	RP_BrakeTorqueCmd	Required	IF_BrakeTorqueCmd	Sender-Receiver
BrakeActuatorManager	PP_ActuatorStatus	Provided	IF_ActuatorStatus	Sender-Receiver
DiagnosticEventHandler	RP_DiagEvent	Required	IF_DiagEvent	Client-Server
PedalInputHandler	PP_PedalRequest	Provided	IF_PedalRequest	Sender-Receiver
BrakeTorqueCoordinator	RP_PedalRequest	Required	IF_PedalRequest	Sender-Receiver

Table 5-2 defines the port interfaces.

Interface	Type	Data Elements / Operations	Description
IF_WheelSpeed	Sender-Receiver	Sig_WheelSpeed_FL, Sig_WheelSpeed_FR, Sig_WheelSpeed_RL, Sig_WheelSpeed_RR	Filtered wheel speeds.
IF_VehicleState	Sender-Receiver	Sig_VehicleSpeed, Sig_LongitudinalSlip	Estimated vehicle state.
IF_BrakeTorqueCmd	Sender-Receiver	Sig_BrakeTorqueRequest	Target brake torque command.
IF_DiagEvent	Client-Server		Diagnostic event reporting service.
IF_ActuatorStatus	Sender-Receiver	Sig_ActuatorHealth	Actuator health status.
IF_PedalRequest	Sender-Receiver	Sig_BrakePedalPosition	Driver brake request.

6 Signals

Signal	Interface	Data Type	Unit	Range	Producer	Consumer
Sig_WheelSpeed_FL	IF_WheelSpeed	uint16	rpm	0..3000	WheelSpeedProcessor	VehicleStateEstimator
Sig_WheelSpeed_FR	IF_WheelSpeed	uint16	rpm	0..3000	WheelSpeedProcessor	VehicleStateEstimator
Sig_WheelSpeed_RL	IF_WheelSpeed	uint16	rpm	0..3000	WheelSpeedProcessor	VehicleStateEstimator
Sig_WheelSpeed_RR	IF_WheelSpeed	uint16	rpm	0..3000	WheelSpeedProcessor	VehicleStateEstimator
Sig_VehicleSpeed	IF_VehicleState	uint16	0.01 km/h	0..30000	VehicleStateEstimator	BrakeTorqueCoordinator
Sig_LongitudinalSlip	IF_VehicleState	sint8	%	-100..100	VehicleStateEstimator	BrakeTorqueCoordinator
Sig_BrakeTorqueRequest	IF_BrakeTorqueCmd	float32	Nm	0..6000	BrakeTorqueCoordinator	BrakeActuatorManager
Sig_ActuatorHealth	IF_ActuatorStatus	uint8	-	0..3	BrakeActuatorManager	DiagnosticEventHandler
Sig_BrakePedalPosition	IF_PedalRequest	uint8	%	0..100	PedalInputHandler	BrakeTorqueCoordinator

7 Component Dependencies

Table 7-1 lists the documented dependencies between software components.

Source	Relation	Target	Via	Description
VehicleStateEstimator	depends on	WheelSpeedProcessor	IF_WheelSpeed	Needs filtered wheel speeds.
BrakeTorqueCoordinator	depends on	VehicleStateEstimator	IF_VehicleState	Needs vehicle speed and slip.
BrakeActuatorManager	depends on	BrakeTorqueCoordinator	IF_BrakeTorqueCmd	Executes the torque command.
BrakeTorqueCoordinator	depends on	PedalInputHandler	IF_PedalRequest	Needs the driver brake request.
DiagnosticEventHandler	depends on	BrakeActuatorManager	IF_ActuatorStatus	Monitors actuator health.

8 Functional Flows

FF-01 Normal service braking: PedalInputHandler -> BrakeTorqueCoordinator -> BrakeActuatorManager

When the driver presses the brake pedal, the PedalInputHandler publishes Sig_BrakePedalPosition. The BrakeTorqueCoordinator converts the request into Sig_BrakeTorqueRequest using Sig_VehicleSpeed, and the BrakeActuatorManager applies the corresponding valve commands.

FF-02 Wheel slip control: WheelSpeedProcessor -> VehicleStateEstimator -> BrakeTorqueCoordinator -> BrakeActuatorManager

If Sig_LongitudinalSlip exceeds the slip threshold, the BrakeTorqueCoordinator reduces the torque request on the affected axle until the slip returns below the threshold.

FF-03 Fault reporting: BrakeActuatorManager -> DiagnosticEventHandler

Actuator faults are reported to the DiagnosticEventHandler, which stores the event and informs the diagnostic stack.

9 Degraded Mode Handling

If Sig_VehicleSpeed is invalid for more than 100 ms, the BrakeTorqueCoordinator switches to a degraded mode that applies the driver request without slip control. The DiagnosticEventHandler records the transition.

Appendix A Component Catalogue

The catalogue below summarises component purposes for onboarding.

Component	Responsibility
VehicleStateEstimator	Calculates the steering angle target for lane keeping.
BrakeActuatorManager	Commands the hydraulic modulator valves to realise the target brake torque.